

Assignments on Measurement and Instrumentation

***Instruction, First determine the value for X and Y from the third and second last two digits of your NRIC/identity number; e.g. for NRIC number 881011130600. X = 6 and Y = 0.**

8 8 1 0 1 1 1 3 0

6	0
↓	↓
X	Y

 0

$$XY = 60$$

$$X + Y = 6 + 0 = 6$$

$$X \times Y = 6 \times 0 = 0$$

$$X = 6$$

$$Y = 0$$

Assignment 1

You measure a physical quantity and want to determine the error in your measurement. The true value is known to be 100.0 units, but your instrument reading is $100.0 + \frac{XY}{10}$ units (use the third and second last two digits of your NRIC number; e.g. for NRIC number 881011130600, the reading is $100.0 + 6.0 = 106.0$ units).

Calculate the absolute error (measured minus true value) and the percentage error of this measurement. Express the percentage uncertainty to an appropriate number of significant figures.

Assignment 2

Two resistors R and Rs are connected in series. The voltage drops across each resistor are measured respectively as,

$$E = 10 \text{ V} \pm 0.1 \text{ V (1 \%)}$$

$$E_s = 1.2 \text{ V} \pm 0.0XY \text{ V}$$

Along with a value of $R_s = 0.0XY6 \Omega \pm 0.25\%$

Determine the power dissipated in resistor R and its uncertainty.

Replace the digit **XY** with the third and second last digits of your matric number. use the third and second last two digits of your NRIC number; e.g. for NRIC number 881011130600, $E_s = 1.2 \text{ V} \pm 0.060 \text{ V}$

Assignment 3

Two resistors are measured and then connected in series. Let $R_1 = (100 + (\mathbf{X} + \mathbf{Y}) \times 2)\Omega$ and $R_2 = (50 + (\mathbf{X} + \mathbf{Y}) \times 5)\Omega$. Each has an uncertainty of $\pm 5\%$ in its measured value. Using the rules for uncertainty in additive quantities, first compute the total series resistance $R_{\text{total}} = R_1 + R_2$, and then determine the uncertainty in R_{total} . (Assume the uncertainties in R_1 and R_2 are independent; combine them in quadrature for the sum.)

Assignment 4

To assess measurement precision, you will calculate the standard deviation of a set of readings. Consider four measurements of a length (in cm):

$$L_1 = 10.0 + 0.X$$

$$L_2 = 10.2 + 0.Y$$

$$L_3 = 9.8 + 0.X$$

$$L_4 = 10.1 + 0.Y.$$

Compute the mean and the standard deviation of these four values, showing each step.

Assignment 5

A Resistance Temperature Detector (RTD) has a nominal formula

$$R(T) = R_0[1 + \alpha(T - 20^\circ\text{C})].$$

Let $R_0 = (100 + \mathbf{XY})\Omega \pm 1\%$ be the resistance at 20°C ,

$\alpha = 0.004^\circ\text{C}^{-1} \pm 1\%$ the temperature coefficient, and assume the temperature is

$$T = 20 + (\mathbf{Y} + 1) \times 5^\circ\text{C} \pm 0.5^\circ\text{C}.$$

Calculate the resistance $R(T)$ at this temperature, and then estimate the uncertainty in R by propagating the uncertainties in R_0 , α , and T .

(Hint: Use partial derivatives or percentage uncertainties for each factor; quote the final uncertainty in ohms.)

Assignment 6

An analog PMMC voltmeter can be converted to a higher range by adding a series resistor (multiplier). Suppose you have a basic galvanometer with an internal resistance

$R_g = 100$ and full-scale current $I_{fs} = 1$ mA.

You want to design a voltmeter that reads up to $V_{max} = (XY + 20)$ V (use your XY digit values) *. Calculate the required series resistor R_{series} that must be added in series with the meter so that full-scale deflection corresponds to V_{max} . (Use the formula $R_{series} = \frac{V_{max}}{I_{fs}} - R_g$, and show the calculation.)

Assignment 7

You are examining an N -bit analog-to-digital converter. Let $N = 8$ bits for a standard ADC, and assume a full-scale input range of $(XY/10+10)$ V (use your **XY digit values**)*. **Determine the resolution** of the ADC in volts per step.

Assignment 8

An oscilloscope can be used to measure signal frequency by observing the waveform's period. Assume the oscilloscope's time base is set to $((X+Y) \times 2 + 1)$ **ms/div** and one full cycle of the signal spans $((X+Y)+2)$ horizontal divisions on the screen. **Calculate the signal's frequency** in Hz. (First find the period: period = time base $\times$ divisions; then $f = 1/\text{period}$. Show your work and final answer in a grid (adjust the size accordingly) as an example below.)

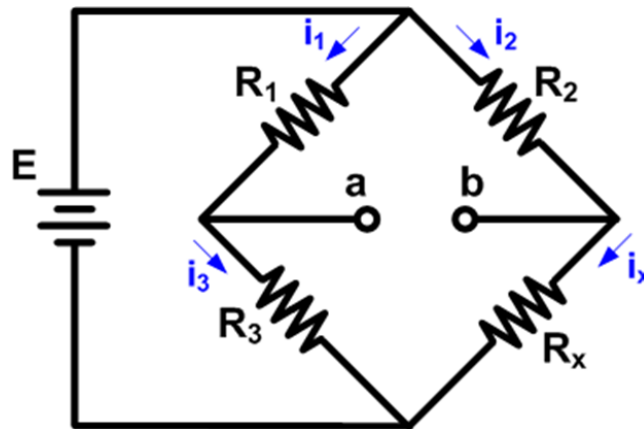


Assignment 9

You have a simple voltage divider with resistors R_1 and R_2 in series across a DC source of 150 V. Let $R_1 = 250 \text{ k}\Omega$ and $R_2 = 2((X+Y) + 1) \text{ k}\Omega$. You connect a voltmeter across R_2 to measure its voltage. The voltmeter has a sensitivity of **20 k Ω /V** and is set to the 150 V range (so internal resistance $R_m \approx 3.0 \text{ M}\Omega$). **Determine the reading of the voltmeter**, i.e. the actual voltage it indicates across R_2 . (Hint: Treat the meter's resistance R_m in parallel with R_2 , find the equivalent resistance across the lower arm, then use voltage division to find the loaded voltage. Briefly show the steps.)

Assignment 10

A **Wheatstone Bridge** is set up with three known resistors and one unknown R_x . In our scenario, let $R_1 = 50 \text{ }\Omega$, $R_2 = 100 \text{ }\Omega$, and $R_3 = (50 + \text{XY}) \text{ }\Omega$. The bridge is balanced (null detector shows zero) when these values are used. **Calculate the unknown resistance R_x** using the Wheatstone balance formula $R_1/R_2 = R_3/R_x$. (Show the formula rearranged to $R_x = \frac{R_3 \times R_2}{R_1}$ and plug in the numbers to find R_x .)



Assignment 11

A certain thermocouple type exhibits a **linear calibration** from 0 °C to 500 °C, producing **18.68 mV at 500 °C** (with the cold junction at 0 °C). Assume this linear relation holds (approximately true for many thermocouples over limited ranges). If your thermocouple's measured output is $\frac{\text{XY}+1}{10} \text{ mV}$, determine the temperature of the hot junction. **Calculate the corresponding temperature in °C**. (Hint: First find the sensitivity in mV/°C from the full-scale data, then use $T = \frac{V_{\text{meas}}}{V_{500^\circ}} \times 500^\circ\text{C}$. Show your calculation.)

Assignment 12

You are analyzing a parallel-plate **capacitive sensor** (e.g. a pressure diaphragm capacitor). The capacitance is given by $C = \frac{\epsilon_0 \epsilon_r A}{d}$. Assume the sensor has a plate **area**

$$A = (XY + 10) \times 10^{-4} \text{ m}^2$$

and plate separation $d = 1.0 \times 10^{-3} \text{ m}$. The dielectric between plates is air with $\epsilon_r = 1.0$ (and $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$). **Compute the capacitance C** in farads. (Plug in the values: $C = 8.854 \times 10^{-12} \times 1.0 \times A/d$. Give your result in scientific notation and also as a value in picofarads, pF.)

Assignment 13

A **Linear Variable Differential Transformer (LVDT)** produces an output voltage proportional to core displacement within its linear range. Suppose an LVDT has a sensitivity of **2.00 V/mm**. If the core is displaced by $((X+Y) + 1) \times 0.5 \text{ mm}$ from the null (center) position, **determine the output voltage** from the LVDT. (Multiply the displacement in mm by 2.00 V/mm to get the voltage in volts. State the result, and note whether this is the peak output assuming the given sensitivity. You may assume linearity around the null position for this small displacement.)

Evelyn Goh Yuan Qi (A23C50222)

X: 6, Y: 0

Assignment 1

Instrument reading = $100 + 6 = 106.0$ units

Absolute error = $106 - 100 = 6.0$ units

Percentage error = $\frac{6.0}{100.0} \times 100\%$

= 6.0%

Assignment 2

$E = 10 \pm 0.1$ V

$E_s = 1.2 \pm 0.06$ V

$R_s = 0.0606 \Omega \pm 0.25\%$

$$W_{R_s} = \frac{0.25}{100} \times 0.0606 \\ = 1.515 \times 10^{-4}$$

$$P = \frac{E \times E_s}{R_s} = \frac{10 \times 1.2}{0.0606} \\ = 198.02 \text{ W}$$

$$P \pm W_P = (E' \pm W_E)(E_s' \pm W_{E_s})(R_s^{-1} \pm W_{R_s})$$

$$\frac{W_P}{P} = \sqrt{\left(\frac{W_E \cdot E_s}{E}\right)^2 + \left(\frac{W_{E_s} \cdot E}{E_s}\right)^2 + \left(\frac{W_{R_s} \cdot P}{R_s}\right)^2} \\ = \sqrt{\left(\frac{1 \times 0.1}{10}\right)^2 + \left(\frac{1 \times 0.06}{1.2}\right)^2 + \left(\frac{-1 \times 1.515 \times 10^{-4}}{0.0606}\right)^2} \\ \frac{W_P}{P} = 0.0511$$

$$W_P = 0.0511 \times 198.02$$

$$W_P = 10.12 \text{ W}$$

$$P = 198.02 \text{ W} \pm 10.12 \text{ W}$$

Assignment 3

$$R_1 = 100 \pm 6(2)$$

$$R_1 = 112 \Omega$$

$$R_2 = 50 \pm 6(5) = 80 \Omega$$

$$R_{\text{total}} = 112 + 80 = 192 \Omega$$

$$W_{R_1} = 0.05(112) = 5.6$$

$$W_{R_2} = 0.05(80) = 4.0$$

$$W_{R_{\text{total}}} = \sqrt{5.6^2 + 4.0^2} \\ = 6.88 \Omega$$

$$R_{\text{total}} = 192 \pm 6.88 \Omega$$

Assignment 4

$$L_1 = 10.0 + 0.6 = 10.6$$

$$L_2 = 10.2 + 0.0 = 10.2$$

$$L_3 = 9.8 + 0.6 = 10.4$$

$$L_4 = 10.1 + 0.0 = 10.1$$

$$\text{Mean} = \frac{10.6 + 10.2 + 10.4 + 10.1}{4}$$

$$= 10.325 \text{ cm}$$

	x	$dx = x - \bar{x}$	dx^2
L_1	10.6	0.275	0.075625
L_2	10.2	-0.125	0.015625
L_3	10.4	0.075	0.005625
L_4	10.1	-0.225	0.050625
			0.1475

$$\text{Standard deviation} = \sqrt{\frac{0.1475}{4-1}} = 0.2217 \text{ cm}$$

Assignment 5

$$R = R_0 [1 + \alpha(T - T_0)]$$

$$R_0 = 100 + 60 = 160 \Omega \pm 1\%$$

$$T = 20 + (0 \pm 1)5 = 25^\circ\text{C} \pm 0.5$$

$$\alpha = 0.004 \pm 1\%$$

$$R = 160 [1 + 0.004 (25 - 20)]$$

$$R = 163.2 \Omega$$

$$\begin{aligned}\frac{\partial R}{\partial R_0} &= 1 + \alpha(T - T_0) \\ &= 1 + 0.004(25 - 20) \\ &= 1.02\end{aligned}$$

$$\begin{aligned}\frac{\partial R}{\partial \alpha} &= R_0 (T - T_0) \\ &= 160(25 - 20) \\ &= 800\end{aligned}$$

$$\begin{aligned}WR &= \sqrt{\left(\frac{\partial R}{\partial R_0} \times WR_0\right)^2 + \left(\frac{\partial R}{\partial \alpha} \times W_\alpha\right)^2 + \left(\frac{\partial R}{\partial T} \times W_T\right)^2} \\ &= \sqrt{(1.02 \times 1.6)^2 + (800 \times 4 \times 10^{-5})^2 + (0.64 \times 0.5)^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial R}{\partial T} &= \alpha R_0 \\ &= 0.004 \times 160 \\ &= 0.64\end{aligned}$$

$$WR = \pm 1.66 \Omega$$

$$\therefore R = 163.2 \pm 1.66 \Omega$$

Assignment 6

$$\begin{aligned} V_{\max} &= XY + 20 & I_{fs} &= 1 \text{ mA} \\ &= 60 + 20 & R_g &= 100 \Omega \\ &= 80 \text{ V} \end{aligned}$$

$$\begin{aligned} R_{\text{series}} &= \frac{V_{\max}}{I_{fs}} - R_g \\ &= \frac{80}{1 \times 10^{-3}} - 100 \\ &= 79900 \\ &= 79.9 \text{ k}\Omega \end{aligned}$$

Assignment 7

$$\begin{aligned} \text{Full scale range} &= \frac{XY}{10} + 10 & N &= 8 \\ &= \frac{60}{10} + 10 \\ &= 6 + 10 \\ &= 16 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Resolution} &= \frac{16}{2^8} \\ &= 0.0625 \text{ V/step} \end{aligned}$$

Assignment 8

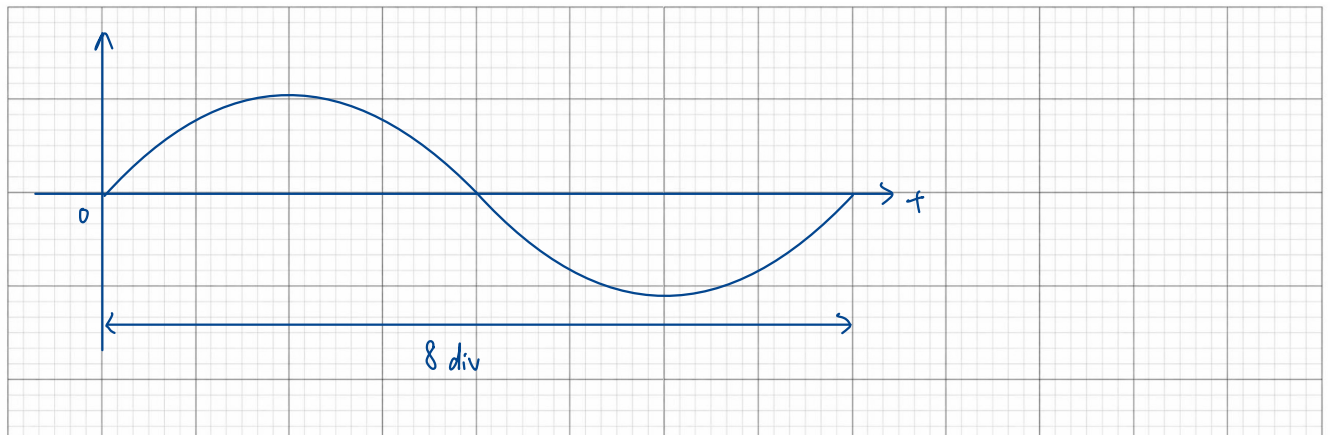
$$\begin{aligned} \text{Time base: } (X+Y)2+1 &= 6(2)+1 \\ &= 13 \text{ ms/div} \end{aligned}$$

$$\text{Divisions per cycle: } (X+Y)+2 = 6+2 = 8$$

$$\begin{aligned} \text{Period} &= 13 \text{ ms/div} \times 8 \\ &= 104 \text{ ms} \\ &= 0.104 \text{ s} \end{aligned}$$

$$f = \frac{1}{0.104}$$

$$f = 9.6214 \text{ Hz}$$



Assignment 9

$$R_1 = 250 \text{ k}\Omega$$

$$R_2 = 2((6+0) + 1) = 14 \text{ k}\Omega$$

$$\text{Sensitivity} = 20 \text{ k}\Omega/\text{V on } 150 \text{ V}$$

$$R_M = 3.0 \text{ M}\Omega$$

$$R_{eq} = R_2 || R_M = \frac{R_2 R_M}{R_2 + R_M} = \frac{14 \text{ k} \times 3000 \text{ k}}{14 \text{ k} + 3000 \text{ k}} \\ = 13.935 \text{ k}\Omega$$

$$V_{R2} = 150 \times \frac{R_{eq}}{R_1 + R_{eq}} \\ = 150 \times \frac{13.935}{250 + 13.935}$$

$$V_{R2} = 7.92 \text{ V}$$

Assignment 10

$$R_1 = 50 \Omega \quad R_2 = 100 \Omega \quad R_3 = 50 + 60 = 110 \Omega$$

$$R_x = \frac{R_3 R_2}{R_1} \\ = \frac{110 \times 100}{50} \\ = 220 \Omega$$

Assignment 11

$$\text{Measured output} = \frac{XY + 1}{10} = \frac{60 + 1}{10} \\ = 6.1 \text{ mV}$$

$$T = \frac{6.1}{18.68} \times 500$$

$$T = 163.28^\circ \text{C}$$

Assignment 12

$$A = (XY + 10) \times 10^{-4} \quad d = 10^{-3} \text{ m}$$

$$A = (60 + 10) \times 10^{-4}$$

$$A = 70 \times 10^{-4} \text{ m}^2$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$= \frac{(8.854 \times 10^{-12})(1)(70 \times 10^{-4})}{1 \times 10^{-3}}$$

$$= 6.198 \times 10^{-11} \text{ F}$$

$$= (6.198 \times 10^{-11} \times 10^{12}) \text{ pF}$$

$$C = 61.98 \text{ pF}$$

Assignment 13

$$\begin{aligned}\text{Displacement} &: ((x+r) + 1) 0.5 \\ &= ((b+0) + 1) 0.5 \\ &= 7(0.5) \\ &= 3.5 \text{ mm}\end{aligned}$$

$$V = 2.00 \times 3.5$$

$$V = 7.0 \text{ V}$$

$\therefore$ Since the LVDT sensitivity is 2.00 V/mm and the displacement from the null position is 3.5 mm , the output voltage is 7.0 V . This value represents the peak output magnitude for that displacement. The polarity depends on the direction of movement from the null.